# Fresh Cart Grocery Performance Dashboard

Project problem statement, business glossary, KPI definitions and analytical scope

## 1. Problem statement

A grocery delivery operation needs a reliable way to track sales performance and service quality across products, delivery zones, customer segments and time periods. Raw order records contain cancellations, inconsistent category or zone labels and incomplete delivery fields. Without a cleaned and reconciled reporting layer, managers cannot consistently measure net revenue, delivery success, fulfilment speed or customer experience. This Excel project organizes raw grocery orders into an auditable cleaned dataset and an interactive performance dashboard.

## 2. Project objectives

- Track net revenue, order volume, average order value and delivered-customer reach.
- Monitor delivery success, on-time performance, delivery and fulfilment durations, and customer ratings.
- Compare performance by zone, category, product, weekday and time band.
- Provide date, zone and category selectors with source-to-dashboard reconciliation.

## 3. Workbook inventory and scope

Seven worksheets: Dashboard (presentation and filters), Analysis (KPI calculations and ranked summaries), Customers (customer-level delivered-order counts), Settings (filter choices), Clean (900 standardized order records and derived fields), Raw (source records), and Audit (reconciliation checks). The dashboard currently selects the complete 30-day data period with all zones and categories. The source owner, collection method, currency, and data licence are not documented.

## 4. Headline dashboard measures

| Field / measure | Meaning / interpretation |
|---|---|
| Net revenue | 310,860.65; currency unspecified |
| Total orders | 900 |
| Delivered orders | 834 |
| Unique delivered customers | 284 |
| Average order value | 372.73 (net revenue / delivered orders) |
| Average delivery minutes | 24.31 |
| On-time delivery rate | 75.06%; confirm the workbook's threshold |
| Delivery success rate | 92.67% (834 / 900) |
| Average customer rating | 4.25 / 5 |

## 5. Data dictionary and terminology

| Field / measure | Meaning / interpretation |
|---|---|
| OrderID | Unique order identifier; validate uniqueness. |
| OrderDate | Date of order; stored as an Excel date serial. |
| OrderHour | Hour of the day the order was placed. |
| CustomerID | Customer identifier; pseudonymous but still potentially linkable. |
| CustomerType | Customer segment, e.g. Returning. |
| Zone | Delivery area; standardized in the Clean sheet. |
| Category | Grocery product category. |
| Product | Product purchased. |
| Quantity | Number of units ordered. |
| UnitPrice | Price per unit before discount; currency not specified. |
| DiscountPct | Discount rate as decimal, e.g. 0.10 = 10%. |
| Status | Order outcome, e.g. Delivered or Cancelled. |
| DeliveryMinutes | Recorded delivery duration; blank for some undelivered orders. |
| FulfilmentMinutes | Recorded order preparation/fulfilment duration. |
| Rating | Customer rating where available. |
| GrossSales | Quantity × UnitPrice. |
| DiscountAmount | GrossSales × DiscountPct. |
| NetRevenue | Realized order revenue after discounts and status rules; cancelled orders may be zero. |
| TimeBand | Derived grouping of order hour, e.g. Afternoon. |
| Weekday | Derived abbreviated day name. |
| WeekdayNo | Numeric weekday sort order. |
| TimeBandNo | Numeric time-band sort order. |

## 6. KPI definitions and calculation cautions

- Total orders: count of cleaned OrderID records. Delivered orders: count with Status = Delivered.
- Net revenue: sum of NetRevenue, applying the workbook's cancellation treatment.
- Average order value: net revenue divided by delivered orders; denominator matters.
- Delivery success rate: delivered orders divided by all orders × 100.
- Unique delivered customers: distinct CustomerID values for delivered orders.
- Average delivery time: mean DeliveryMinutes for applicable delivered orders.
- On-time delivery rate: proportion of qualifying deliveries meeting the workbook's on-time rule; verify threshold in formulas.
- Average rating: mean of recorded nonblank ratings, not all orders.

## 7. Suggested analytical questions

- Which categories and products contribute the most net revenue?
- How do revenue and order counts vary by zone and day?
- Which customer segments and order time bands contribute most to demand?
- Are delivery times, on-time rates and ratings consistent across zones?

- How do cancellations and discounts affect realized revenue?

## 8. Data quality, assumptions and limitations

- The Raw worksheet contains more rows than the 900-record Clean table; review source-to-clean exclusions and Audit checks before claiming complete coverage.
- Check standardized text values, duplicate OrderID values, missing delivery/rating fields and formula outputs.
- The dashboard date range is based on workbook date serials; do not assume a live or continuously updated feed.
- Net revenue is not gross profit; product costs, refunds and overhead are not documented.
- Correlation between delivery time and rating does not establish causation.
- Confirm rights to publish and consider removing CustomerID before releasing raw-level data.

## 9. Tools and deliverables

Microsoft Excel: source cleaning, derived measures, formulas, customer aggregation, interactive dashboard and audit checks. Portfolio deliverables: Excel workbook, project documentation, README.md and optional dashboard screenshot.

## 10. Portfolio description

Built an Excel grocery performance dashboard using 900 cleaned orders to monitor net revenue, deliveries, customer reach, fulfilment efficiency and product performance. Designed date, zone and category filters, KPI calculations and reconciliation checks to support operational reporting.